

Homework (Habanero)

Open Ended Questions (FRQ)

- Q9.** A certain type of cancer cell grows at a rate that is directly proportional to the concentration of a certain protein in the bloodstream. If 500 cancer cells are present when the protein concentration is 10mg/L, and 2000 cancer cells are present after 2 hours, what is the rate of growth of the cancer cells in cells/hour? Show all work and explain your reasoning. Assume the protein concentration remains constant.
- Q10.** A doctor is conducting an experiment to determine the effect of a certain medication on blood pressure. The doctor measures the blood pressure of 10 patients before and after taking the medication, and records the results in the table below. Using the data, calculate the average rate of change of blood pressure in mmHg/hour and explain what this value represents in the context of the experiment.

Chef's Tips

- Chef's Tips
- When solving problems involving rates and variation, be sure to identify the variables and constants in the problem.
- Use units and dimensional analysis to ensure that your answers have the correct units and make sense in the context of the problem.